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Short-term pre-aeration applied to the dry anaerobic digestion of MSW, with a focus on the spectroscopic characteristics of dissolved organic matter

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HIGHLIGHTS

- MSW with an 8-day pre-aeration produced the highest biogas yield of 192.4 NL/kg DM.
- Butyric acid dominated the VFAs (50–55%) in over acidifying reactors during dry AD.
- Accelerated protein degradation by pre-aeration neutralized excessive VFAs.
- Pre-aeration facilitated the conversion of protein- to fulvic- and humic-like matter.
- Fulvic- to protein-like fluorescent matter ratio can evaluate the stability of dry AD.

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ABSTRACT

There is a growing trend toward considering municipal solid waste (MSW) with a large organic fraction as a source of renewable energy using a dry anaerobic digestion (AD) process. Pre-aeration is effective in improving the dry AD of MSW. However, due to a lack of information regarding the modification of organic components during pre-aeration and the relationship with the subsequent AD, the mechanisms and optimum strength of this process remain unclear.

In this study, the role of different short-term pre-aeration durations (4, 8, and 14 days) in dry AD of MSW was evaluated in batch-tests, with a focus on changes in the organic components during pre-aeration and their impact on the subsequent spectral evolution of dissolved organic matter (DOM) and bulk performance. Dry AD of MSW operated effectively after the 8-day pre-aeration. The accelerated degradation of proteins during pre-aeration is critical for the subsequent AD, by reducing the organic loading and providing a buffer for the ammonia generated. A favorable methanogenesis, facilitated by pre-aeration, rapidly decreased protein-like fluorescent matter by ~2.8 times, by converting it to stable humic- and fulvic-like groups. MSW with an 8-day pre-aeration produced the highest biogas yield (192.4 NL/kg dry matter), representing ~24.9% of the initial carbon recovered as methane. Excessive pre-aeration of 14 days decreased the biogas yield by ~33%. An insufficient pre-aeration led to an excessive butyric-type acidification, with protein-like matter dominating the bulk DOM (~81% of total fluorescence intensity). Multivariate analysis methods confirmed that spectroscopic indicators could evaluate the stability of dry AD.

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1. Introduction

The generation of municipal solid waste (MSW) in China has increased due to rapid socioeconomic and population growth. It has been reported that MSW collected in China amounted to ~1.79 million tons in 2014 [1], with an estimated increment of 3–10% per year [2]. MSW generated in developing countries is characterized by a high content of biodegradable organics and moisture, with food waste usually accounting for more than 50% by weight [3,4]. The improper processing and/or disposal of MSW can have serious adverse effects on the surrounding environment. However, the special chemical characteristics of MSW have generated interest in its use as a substrate for methane-rich gas via anaerobic processes, such as dry/wet anaerobic digestion (AD) or anaerobic bioreactor landfilling [5–8]. This could provide a solution

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for the treatment of MSW together with the recovery of energy from the biogas produced to respond to the challenge of global climate change.

A dry batch AD system is an important reactor type due to its relative technical simplicity, the minimal capital cost and energy consumption, and the limited requirements for waste pre-sorting [9]. This process may be used as an in-vessel landfill, but with temperature control and continuous leachate/percolate recirculation, and little or no additional water, or sludge, needed be added to the substrates [10]. However, the dry AD of raw MSW, especially MSW with a large easily degradable fraction, may be especially sensitive to the inhibition caused by excessive acidification due to organic overloading of the system. This will result in low operational stability and hinder the actual application [6,11,12]. To overcome this problem, a large amount of inoculum or hydroxide is usually applied in the early AD phase [8,10,13]. However, there are disadvantages with these approaches because the efficiency of the digester is decreased as some of the volume capacity is lost, due to the presence of inoculum, and the salt content increases dramatically in the digester.

Therefore, to improve the digestion performance in a dry AD system (including MSW bioreactor landfills), a short-term biological pre-aeration (either in-situ or ex-situ aeration mode) prior to anaerobic operation has received much interest in recent studies [5,14–16]. The advantages of a short-term pre-aeration have already been proven in the references mentioned above, and include the rapid development of methanogenesis for energy recycling and saving the volume capacity of digesters. Especially, the process requires relatively little investment cost and process control is straightforward, which makes it particularly attractive for developing countries [4]. Previous studies have suggested using volatile solids (VS) (~15%) and carbon losses (20–30%) as criteria to determine the appropriate time to commence the subsequent anaerobic operation [17,18]. However, the use of these parameters is limited by the lack of information regarding the specific components remaining after pre-aeration, where specific organic components in waste significantly influence its subsequent anaerobic degradation behavior [7]. Due to the lack of a consistent aerobic intensity, previous studies have neglected the underlying correlations between the modification of organic components during the pre-aeration stage and its relationship with the subsequent performance of AD, thereby limiting our understanding of the potential application of this combined process.

Microorganisms only utilize organic matter when they are dissolved in water. Most of the organic compounds in leachate are present within the dissolved organic matter (DOM), including dissolved components and the intermediates of microbial processes during the degradation of organic matter, and can be used to reflect the progress of the anaerobic process (e.g., the hydrolysis, acidification, and methanogenesis phases) [19,20]. Fluorescence excitation-emission matrix (EEM) spectroscopy provides a rapid method to characterize the fractions of DOM in the liquid phase, revealing the different types of fluorescent materials, such as protein-like, fulvic acid-like and humic acid-like substances [21]. Previous studies have demonstrated that this technique, with its sensitivity and selective advantages, could be used to evaluate the properties of water-extractable organic matter and/or DOM during composting, landfilling, and AD processes, with respect to various substrates [22–24]. However, to the best of our knowledge, no study has been conducted to compare the distinction in spectroscopy characteristics during the dry AD of MSW with different pre-aeration levels, and therefore this was the focus of this study. The results will help to explain the degradation mechanisms of pre-aerated MSW in a dry AD process, and thus enable a suitable pre-aeration strength to be determined.

In this study, a series of short-term pre-aeration durations (0, 4, 8, and 14 days) were applied prior to the dry anaerobic degradation of MSW in batch bioreactors. The novel aspect of this study was to compare the effects of three different short-term pre-aeration time periods on the modification of organic components in waste, and its relationship with subsequent AD with respect to the evolution of liquid digestate and biogas generation. Fluorescence EEM spectroscopy coupled with parallel factor (PARAFAC) analysis was applied to investigate the spectroscopic changes of DOM during the AD process. Subsequently, a combination of a Pearson analysis and principal component analysis (PCA) was used to evaluate the inter-relationships between chemical and spectral indicators. The results obtained above were used to discover the mechanisms and potential application of short-term pre-aeration for enhancing the dry AD of MSW.

2. Materials and methods

2.1. Substrate – synthetic MSW

Synthetic MSW was prepared in the lab to simulate the typical composition of MSW generated in Beijing City [25]. In this way, the discrepancy of substrate composition among the bioreactors was minimized, allowing for more comparable results. The synthetic MSW was prepared using a mixture (on a wet weight basis) of food waste (63%), office paper (13%), plastic (5%), and inorganic materials, such as stones, glasses and metals (19%), respectively. The initial water content and biodegradable VS content of the synthetic MSW were 57.5% (w/w) and 47.3% dry matter (DM), respectively. The other chemical components and eluate characteristics from the leaching test are shown in Table 1. Before loading into each bioreactor, the synthetic MSW was manually shredded and well mixed, with a uniform size of 20–40 mm.

2.2. Column experiments

2.2.1. Column equipment

Four laboratory-scale Plexiglas columns were used to compare the effects of different short-term pre-aeration durations on the dry AD of MSW. Each bioreactor had an internal diameter of 180 mm and a height of 320 mm. Detailed information regarding the columns is provided elsewhere [26]. The bioreactor was designed to allow operation under both aerobic and anaerobic conditions by selectively switching the valves on or off. The operational temperature was kept constant at 35 ± 1 °C through a thermo-regulated insulation system, which covered the lateral surfaces of the bioreactors.

2.2.2. Pre-aeration stage

Three kilos of MSW were loaded into each column, except the control (AN bioreactor). While loading, approximately 150 g of sludge compost (5% inoculation rate) was mixed in the waste to enhance the aerobic degradation process. The short-term pre-aeration duration was the sole variable. Three bioreactors were operated with 4, 8, and 14 days pre-aeration, respectively, corresponding to bioreactors R(4), R(8), and R(14). The initial airflow rate was a preset cycle of 10 min aeration followed by a 20 min pause, with an airflow rate of 0.5 L/min/kg waste. The headspace gas composition before and after ventilation was measured. An optimum oxygen level of 5–15% (v/v) was maintained by regulating the ventilation frequency during the whole pre-aeration phase. Additionally, the leachate produced from each bioreactor was drained daily and no leachate recirculation was applied during the pre-aeration stage.

Table 1
Quality parameters of the initial waste and pre-aerated wastes.

	Parameters[a]	Initial waste	Waste after 4-day pre-aeration	Waste after 8-day pre-aeration	Waste after 14-day pre-aeration
Solid phase	Dry matter (DM, %)	42.5	45.7	44.5	44.1
	Biodegradable VS (%DM)	47.3	42.3	33.5	24.8
	Proteins (g/g TS[b])	0.229	0.226	0.152	0.160
	Lipids (g/g TS)	0.155	0.132	0.119	0.114
	Carbohydrates (g/g TS)	0.174	0.094	0.047	0.005
	Lignocelluloses (g/g TS)	0.363	0.337	0.389	0.400
Eluate from leaching test	pH	6.24	5.15	7.03	6.75
	TOC (mg/L)	4227	5777	2275	2687
	VFAs (mg/L)	115.9	681.5	133.7	91.6
	Alkalinity (mg $CaCO_3$/L)	519	1435	1038	916
	NH_4^+-N (mg/L)	51	878	1242	1345
	$SUVA_{254}$ ($\times 10^{-2}$, L/mg/m)	0.4	0.2	1.1	1.2

[a] Parameters were tested after plastic and inorganic materials removed except for DM and biodegradable VS.
[b] TS is the abbreviation of total solid, here representing the solid residuals after plastic and inorganic materials removed.

2.2.3. Dry batch AD stage

Destructive sampling was undertaken when each bioreactor completed its designed pre-aeration term (i.e., 4, 8, and 14 days, respectively). Approximately one tenth (w/w) of the pre-treated waste was sampled using a sequential quartering procedure to guarantee the homogeneity of later analyses. The remaining pre-aerated waste was re-filled into an individual bioreactor. For the AN bioreactor, only 2.5 kg of raw MSW was loaded, with no aerobic compost or pre-aeration applied.

Anaerobically digested sludge, obtained from a full-scale AD plant located in Beijing, was added as seeding material, accounting for 10% (w/w) of the waste loaded in all of the bioreactors (i.e., AN, R(4), R(8), and R(14)). The pH, total solids (TS), and VS of seeding sludge were 7.8, 7.2% (w/w) and 46.5%TS, respectively. Additionally, 200–250 mL of deionized water (~10% weight of the loaded waste) was introduced into each bioreactor for the moisture content adjustment and initial leachate production. After these operations, all of the bioreactors were subjected to anaerobic conditions. The leachate generated from each bioreactor was manually recirculated through a disperser in the top of the bioreactor every 2 days, and tested every 4 days throughout the whole anaerobic period (100 days). The leachate withdrawn for analysis at each sampling (25–30 mL) was replaced by an equivalent volume of deionized water. The biogas generated was collected using 5 L aluminum sampling bags and analyzed every 2 days.

2.3. Characterisation of spectroscopy

Fluorescence EEM spectroscopy was measured using a Hitachi F-7000 fluorescence spectrometer (Hitachi, Japan) at room temperature according to He et al. [23], with minor modifications. The leachate samples were filtered through 0.45 μm membrane filters and were preliminarily diluted to a TOC concentration of ~5 mg/L using deionized water. The slit widths were 5 nm for the excitation and emission monochromators, and the scan speed was set to 12,000 nm/min. EEM spectra were gathered with the subsequent scanning emission spectra (Em) from 240 to 600 nm at 1 nm increments by varying the excitation wavelength (Ex) from 220 to 450 nm at 5 nm increments. The leachate samples extracted from the AD process in all runs on days 28, 56, and 84 were used for EEM spectroscopy characterization. In addition, the fluorescence spectrum for ultrapure water was subtracted from the spectrum of the sampled leachates to eliminate water Raman scattering and reduce background noise.

All EEM spectral data were converted into excel format. A PARAFAC analysis was then used to mathematically deconvolute the original EEM spectral data into different independent groups of fluorescent components. This PARAFAC analysis was performed using MATLAB software (ver. 7.0; MathWorks, Inc., Natick, MA, USA) with the DOMFluor toolbox (www.models.life.ku.dk), in accordance with previous reports [23,24,27]. In detail, after several post-acquisition steps (i.e., removal of scattering and data arrangement) to correct the fluorescence EEM spectra, a series of PARAFAC models with two to seven components were fitted to the test data. A split half analysis (Fig. S1) and residual analysis (Fig. S2) were used to identify the correct number of components. The concentration scores of the PARAFAC components were normalized to the Raman peak area (R.U.) and expressed as the maximum fluorescence intensity (F_{max}).

2.4. Other analytical methods

The physical and chemical characteristics of the solid samples before and after pre-aeration were measured. The moisture content was determined by heating the solid samples at 105 °C. The VS content was determined by ashing the dry samples, after the removal of plastics, at 550 °C to a constant weight in a muffle furnace. Preliminary dried and shredded solid samples (granule scale less than 1 mm) were used to analyze their organic components. Elemental analysis was undertaken using a carbon-hydrogen-nitrogen analyzer (model CE 440; EAI, Oakland, NJ, USA). Proteins, lipids, and lignocellulose content were determined according to a method used in a previous study [28]. The carbohydrate content was calculated by subtracting the sum of lipids, proteins, and lignocellulose content from the VS.

The waste eluates were obtained by the water extraction of pre-dried and ground waste, shaken for 24 h in ultrapure water (1:10 ratio of solid/liquid w/w) at room temperature. Liquid samples, both from waste eluates and anaerobic leachate, were centrifuged at 10,000 for 10 min and filtered through 0.45 μm membrane filters. The filtrate was tested for pH, dissolved organic carbon (DOC), alkalinity, soluble chemical oxygen demand (SCOD), NH_4^+-N, and VFAs according to APHA methods [29]. The concentrations of DOC and VFA were measured using a total organic carbon analyzer (TOC-Vcph, Shimadzu, Kyoto, Japan) and gas chromatography equipped with a flame ionization detector (GC-2014, Shimadzu), respectively. SUVA254 values (reported in L/mg/m) were determined by dividing the UV–Vis absorbance measured at $\lambda = 254$ nm by the TOC concentration [20].

The O_2 emissions during the short-term pre-aeration and the biogas produced during the AD process (mainly CH_4 and CO_2) were measured by gas chromatography using a thermal conductivity detector (GC-2014, Shimadzu).

All analyses of solid, liquid, and gas samples were conducted in duplicate.

2.5. Statistical methods

PCA was conducted to analyze the correlations between chemical and spectroscopic indicators in anaerobic leachate based on their correlation matrix, with a dataset normalized by taking the log10, extracting the mean, and dividing it by the standard deviation. PCA was performed using CANOCO for Windows software (ver. 4.5; Biometris, Wageningen, Netherlands).

3. Results and discussion

3.1. Characterization of pre-aerated waste

The characteristics of the solid phase and eluate of the initial and pre-aerated waste are given in Table 1.

The pre-aeration process clearly altered the VS content and fractional content of organic components in substrates. The VS content in solid waste decreased by 10.6, 29.2, and 47.6% (based on dry weight) after 4, 8, and 14 days pre-aeration, respectively. Carbohydrates contributed to most of the VS losses during this process, ranging from 29.2 to 52% across the different pre-aeration durations. A clear decrease in the fractional content of proteins was observed between the R(4) and (R8) bioreactors, suggesting that the organic nitrogen degradation mainly occurred over days 5–8, which corresponded to the thermophilic period during the pre-aeration process (average 51.6 °C). In contrast, the relative fraction content of lignocelluloses increased from the initial value of ~0.363 g/g TS to ~0.389 and 0.400 g/g TS in the R(8) and R (14) bioreactors, respectively. This could be explained by the slower degradation rate and special microorganism metabolism pathways required compared with other organic components as reported by Zhang et al. [30], who concluded that inoculation and a sufficient incubation time were indispensable for lignocellulose disintegration and degradation. These findings showed that the degradation of different organic components in waste during the pre-aeration stage was not synchronous.

The results of the leaching test on solid samples revealed significant discrepancies in all parameters. Higher pH values were observed in the samples from bioreactors R(8) and R(14), which were consistent with the lower values of the VFA to alkalinity ratio. The NH_4^+-N concentration increased as the pre-aeration duration was prolonged, while the DOC and VFA concentrations in the leachate extracted from the solid sample in R(4) were clearly higher than those obtained for the other indicators, largely due to the enhanced hydrolysis and acidification achieved by the low-intensity pre-aeration process [31]. Consequently, the pre-aeration periods of 8 and 14 days resulted in decreased DOC and VFA concentrations in the eluate of solid wastes, which was attributed to the final degradation of easily biodegradable organic matter, especially carbohydrates and proteins (Table 1). He et al. [32] reported that the DOC of MSW excavated from an actual landfill is usually less than 1.0 g/kg waste, which is clearly much lower than the result obtained even for the 14-day pre-aerated waste in this present study (~26.9 g/kg waste). These results suggested that even after short-term pre-aeration (e.g., 4–14 days) the wastes were still unstable in terms of DOC.

$SUVA_{254}$ is an indicator that indirectly reflects the degree of biological degradability of the samples. The $SUVA_{254}$ values were increased by ~3 times following 8 and 14 days pre-aeration. Shao et al. [33] observed that the $SUVA_{254}$ values of water extracts of MSWs after the hydrolysis-aerobic process (100 days) were around 3.0 L/mg/m, which was clearly higher than the values from short-term pre-aerated waste in this study, implying a lack of biological stabilization of pre-treated substrates. This finding was consistent with the results for the DOC concentration.

3.2. Bulk characterization of pH, SCOD, NH_4^+-N, VFAs and biogas production during the AD process

The anaerobic phase was operated for a period of around 100 days. Leachate NH_4^+-N, pH, and SCOD concentrations obtained on specific days during the anaerobic landfilling process are shown in Fig. S3. The pH increased in all runs, which could be attributed to the alkaline yield due to the mineralization of proteinaceous materials. A higher pH was observed in bioreactors R(8) and R(14), with final values of 8.2 and 7.9, respectively, at the end of study. The pH of bioreactors AN and R(4) (~6.0) was below the optimum value for methane production throughout the whole experimental period [6,12]. Additionally, pre-aeration clearly increased the initial NH_4^+-N concentration in leachates (samples on day 4 from bioreactors R(4), R(8), and R(14)), which corresponded to the higher initial pH values in individual bioreactors. This result suggested that the increased ammonia levels obtained by pre-aeration could neutralize the VFAs generated during anaerobic treatment, which is beneficial for the development of methanogenesis conditions. The longer the duration of pre-aeration, the lower the final NH_4^+-N concentration that was observed. The lower SCOD for bioreactors R(8) to R(14) suggested that the successful conversion of DOM in leachates was achieved by an 8 and 14 day pre-aeration, with SCOD removal efficiencies of 90.3 and 83.2% at the end of the experiment, respectively. In contrast, the low removal efficiencies of SCOD in the AN and R(4) bioreactors implied that the DOM in leachates was still unstable.

The changes in the VFA content during the AD process in all runs are presented in Fig. 1. An obvious accumulation of VFAs was observed in the AN and R(4) bioreactors throughout the whole experimental period, with similar chemical profiles mainly consisting of butyric acid (50–55%), followed by acetic acid (17–25%) and propionic acid (5–10%), which was typically butyric-type acidification [34]. At the end of study (on day 100), the total VFA content in terms of COD were ~32,000 and ~39,000 mg/L in AN and R(4), respectively. These results implied that the reactors remained in the acidogenesis phase. However, the VFA concentrations in bioreactors R(8) and R(14) decreased rapidly after the initial peak on day 12. Only acetic acid was detected in the final samples, and the average content was less than 20 mg/L, which represents a high AD efficiency.

The biogas production from the AN, R(4), R(8), and R(14) bioreactors during the dry AD process is shown in Fig. S4. There were considerable changes in biogas yield among the bioreactors with the different pre-aeration durations. In bioreactors R(8) and R (14), after the short start-up period (~8 days), methane was the main component in the biogases generated, being present mostly in the range of 50–68%. The accumulated biogas production from bioreactors R(8) and R(14) was 192.4 and 151.6 NL/kg DM, with an average methane content of 56.1 and 55.1% (v/v), respectively. The generation of biogas was almost completed within 54–62 days, suggesting that an accelerated stabilization processes was achieved through an 8- and 14-day pre-aeration. In contrast, there was almost no methane (average content <0.2%, v/v) generated in the AN bioreactor, which was attributable to the excessive acidification, generating only a small amount of biogas consisting of mainly CO_2 and H_2. In the R(4) bioreactor, the accumulated biogas production was 25.0 NL/kg DM at the end of the experimental period. Methane was detected >40% (~40.1%, v/v) for the first time after an anaerobic cultivation of 88 days. Xu et al. [12] reported that methane content greater than 40% is required for the biogas to be technically or economically feasible. This showed that a very long lag-phase occurred in R(4) bioreactor, implying that there was still an overload of organic matter after 4 days of pre-aeration.

All of the regular indicators detected above were in a reasonable range for the anaerobic degradation of real MSW [5,6].

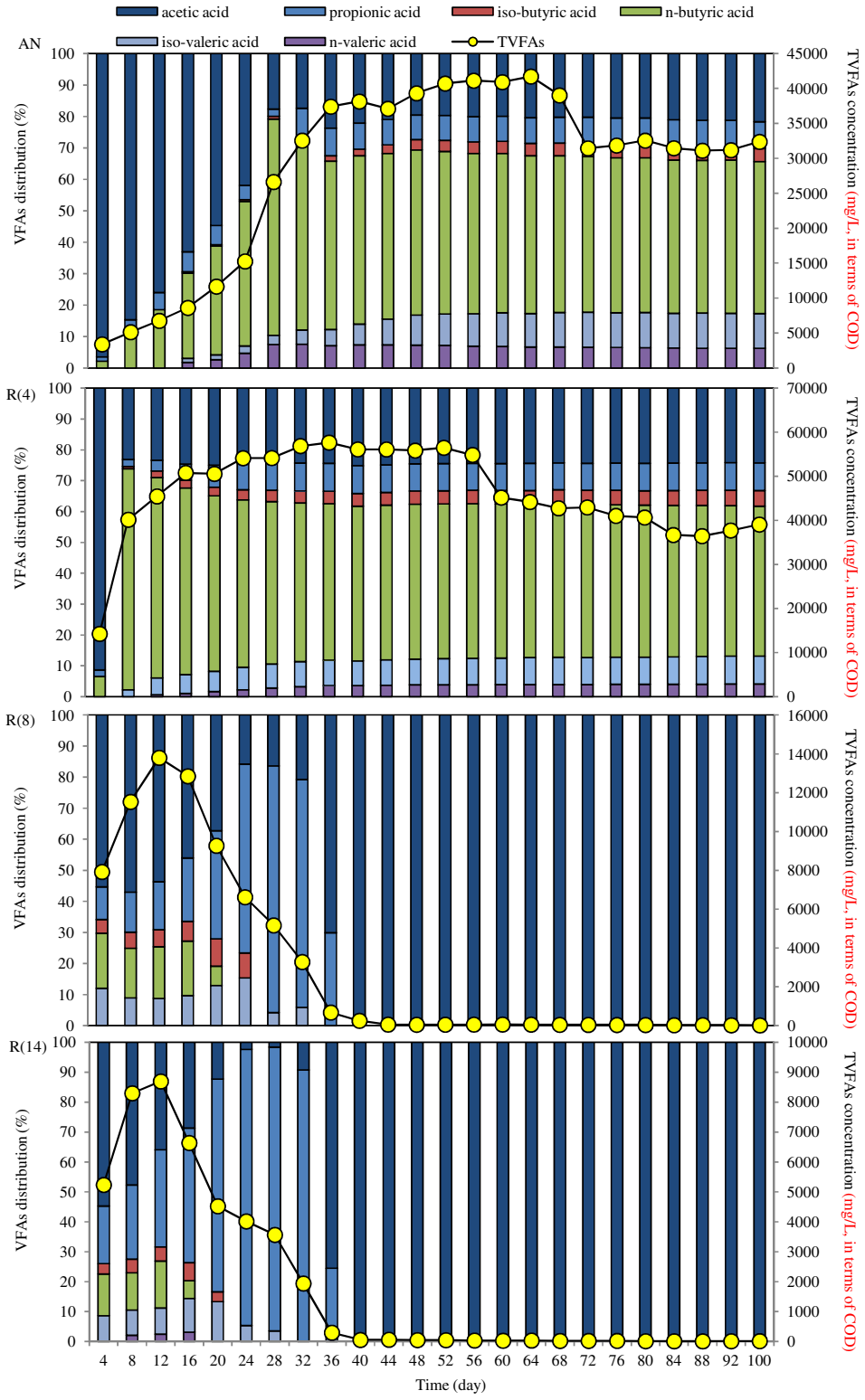


Fig. 1. Evolution of VFAs concentrations in each bioreactor during dry AD process.

3.3. Carbon mass balance combining pre-aeration and AD process

The carbon conversion was evaluated throughout the whole experimental period in all of the bioreactors (Table 2). The analysis was conducted following a method reported by Zhang et al. [35]. The distribution of carbon mass was distinctly different among the bioreactors with different pre-aeration durations. As the pre-aeration duration increased, more organic carbon was converted into the gas phase as CO_2 during this stage (7.6–51.0%), while the carbon released via leachate slightly changed (4.5–7.7%). When considering the combined process, the maximum conversion of carbon into methane (24.9%) was observed in bioreactor R(8), in which the raw waste was pre-aerated for 8 days. The excessive pre-aeration used in bioreactor R(14) further consumed the organic carbon in

Table 2
Carbon conversion during the combined process of pre-aeration and dry AD.[a]

Bioreactor	Initial content(%)	Aerobic stage (%)			Anaerobic digestion stage (%) (at the end of anaerobic operation of 100 days)				Missing (%)
		Gas emissions (calculated[b])	Leachate emissions	Final solid residues	CH_4 emissions	CO_2 emissions	Leachate emissions	C-sink	
AN	100	–			0.03	6.9	8.7	70.8	13.6
R(4)	100	7.6	4.5	87.9	0.4	1.9	5.0	69.8	10.8
R(8)	100	30.0	6.6	63.4	24.9	17.9	0.6	18.0	2.0
R(14)	100	51.0	7.7	41.3	16.6	10.3	0.4	19.3	−5.3

[a] All values are expressed as percentages respect to initial carbon content in the substrate before pre-aeration.
[b] Gas emissions: $initial_{carbon}$ - $carbon_{leachate}$ - $carbon_{residue}$ in the pre-aeration stage.

waste, thereby lowering the methane recovery efficiency, with a decrease of ~33%. The final mass of carbon in solid residuals and leachate was almost the same from bioreactors R(8) and R(14), implying the complete stabilization of organic carbon in these bioreactors. On the other hand, for the AN and R(4) bioreactors, only 15.6 and 19.4% of the initial carbon was released throughout the whole experiment period, respectively, mainly as CO_2 and DOM in leachate. Therefore, ~70% of the initial mass of carbon was left in the solid residues, indicating a very low efficiency of organic mineralization. In conclusions, when considering energy recovery, the duration of pre-aeration prior to anaerobic operation should be around 8 days for MSW with a high organic fraction.

3.4. Spectroscopic characterization of DOM during the AD process

3.4.1. Fluorescence EEM spectroscopy

Fluorescence EEM spectroscopy was used to characterize the DOM in leachates from all of the bioreactors and their spectroscopy priorities over time (i.e., 28-, 56-, and 84-day samples, respectively) are shown in Fig. S5.

According to Chen et al. [21], EEM spectroscopy can be used for the analysis of five regions using consistent excitation and emission wavelength boundaries. A total of five distinct peaks were directly observed during the whole AD process in all runs. Peaks A, B, and C were located in Ex/Em wavelengths of 275/298–310, 275/338–342 and 220–225/305–312 nm, respectively. Both peaks A and B were microbial by-products (i.e., tyrosine- and protein-like, and tryptophan- and protein-like), while, peak C was associated with tyrosine-like materials as free molecules or bound to amino acids and proteins [32]. Peaks D and E are located in Ex/Em wavelengths of 336/430 nm and 280/430 nm, respectively. Previous studies have implied that these two peaks were both attributable to humic-like organic substances [21]. In general, tyrosine-like and tryptophan-like materials are considered to be easily biodegradable substrates [22,33], while continuous mineralization, re-polymerization, and condensation pathways from low-weight molecules or the degradation of lignin lead to the formation of humic-like organic substances, which are effective indicators of organic matter stabilization [23,24].

The spectra obtained from bioreactors AN and R(4) were very similar. Only peaks A, B, and C were directly observed during the whole experimental period (i.e., 28, 56, 84-day samples). These peaks were also found in young leachate sources in a previous study [36]. In bioreactors R(8) and R(14) peaks A, B, and C almost vanished as anaerobic landfilling progressed, while peaks D and E were visually enhanced as anaerobic degradation proceeded (i.e., 56 and 84-day samples), suggesting the stabilization of organic matter in the leachate. The fluorescence EEM spectroscopy results were consistent with the previous results for the bulk characteristics of leachates and biogases in individual bioreactors (Figs. 2, S3 and S4).

3.4.2. Characterization of EEM-PAPAFAC

To acquire more useful information on the evolution of the chemically relevant fluorescent components in DOM, a PAPAFAC analysis was conducted using the raw EEM spectra. The inspection results (Figs. S1 and S2) and visual recognition (Fig. 2) confirmed that the five-component model was suitable for characterizing the DOM extracted from the leachate during the AD process. The maximum Ex/Em wavelength of component C1 was the same as peaks D and E in the raw EEM spectroscopy results, while, that of component C2 was consistent with the raw peaks A and C (Fig. S5). This proved that components C1 and C2 were humic-like and protein-like substances, respectively. Component C3 had two excitation peaks at 225 and 285 nm, with an emission maximum at 385. Zheng et al. [20] indicated that peaks at the Ex/Em wavelength of 230 and 290/390 nm could be attributed to fulvic acid-like components, and therefore C3 was considered to be similar to fulvic acid-like components. Components C4 (Ex/Em wavelength = 250 and 275/440 nm) and C5 (Ex/Em wavelength = 235 and 330/425 nm) had similar fluorescence positions. Compared with component C5, the primary fluorescence peak for component C4 was red shifted, with longer wavelengths attributed to the increase of carbonyl, hydroxyl, alkoxyl, and amino groups in fluorophores [37,38]. Thus, C4 was considered to be similar to fulvic acid-like compounds, while C5 was ascribed to humic-like organic substances [21]. The results of the PARAFAC analysis not only confirmed the previous findings through the direct visual identification of peaks in the raw EEM spectroscopy (components C1 and C2), but also revealed some new peaks which were previously unidentified (components C3, C4, and C5).

3.4.3. Evaluation of spectra changes during AD process

The changes of the concentration scores of the PARAFAC components (F_{max}) reflect the conformation of the fluorescence components under the different states of AD [23,38]. In this study, the changes of the PARAFAC components, especially component C2, could be used to categorize all of the samples into two groups (Fig. 3(a) and (b)). The F_{max} values (0.20–0.26 R.U.) and the relative percentage (>81%) of component C2 from bioreactors AN and R(4) were relatively stable during the whole AD process. In contrast, in the samples from bioreactors R(8) and R(14) there was a clear decrease in the C2 values as AD proceeded, with an average reduction of 2.8 times from the 28 to 84-d samples; meanwhile, for components C1, C4, and C5 there was a significant increase over time. These findings implied that excessive acidification tended to result in a fluorescence peak associated with protein-like matter in leachate. In contrast, a favorable AD process can efficiently decompose protein-like substances, while converting them into more stable fulvic and humic substances.

The ratio of the F_{max} scores of fulvic- and humic-like substances to that of protein-like matter can be used to determine the biodegradability and degree of stabilization of organic matter

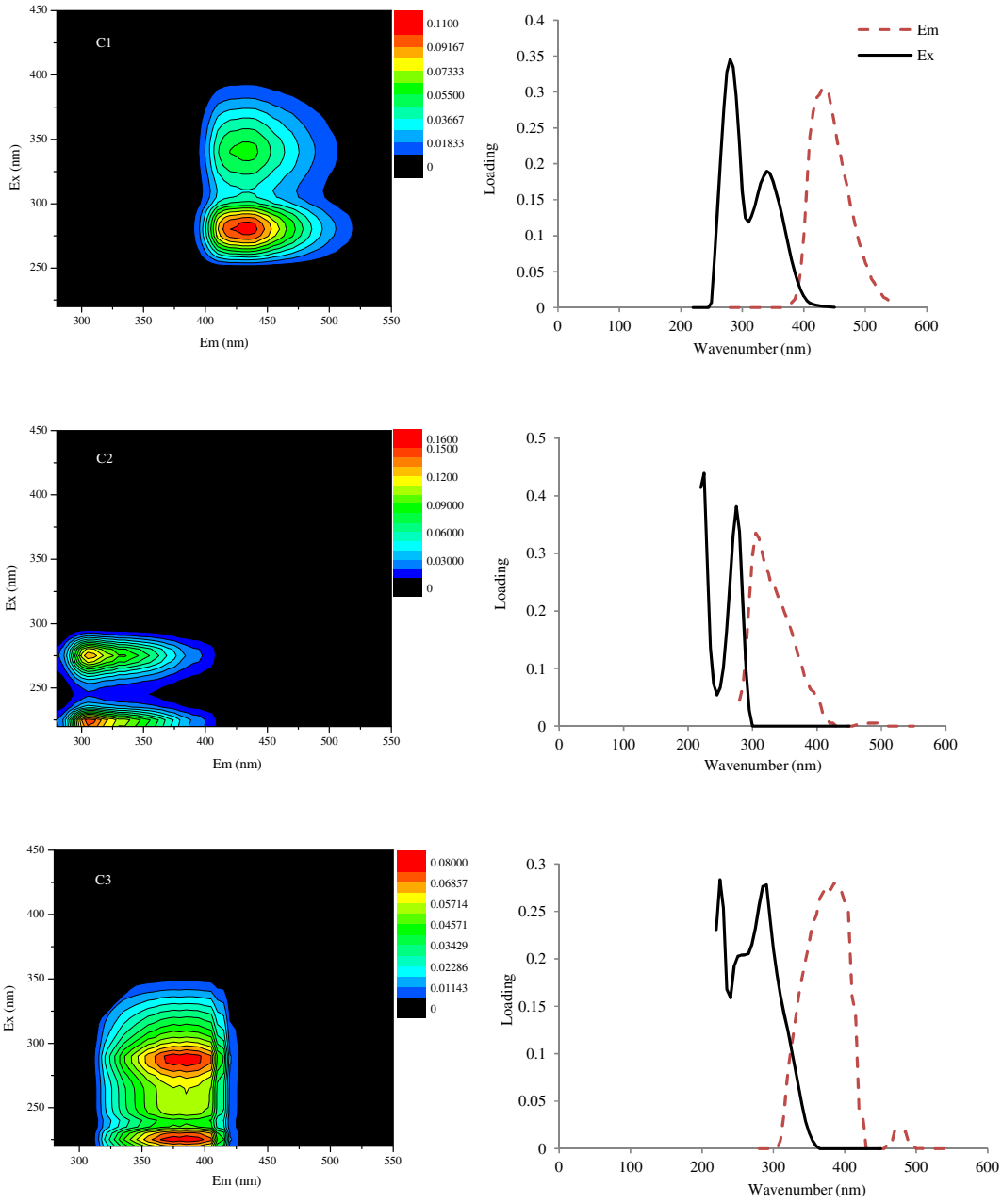


Fig. 2. Five fluorescent components decomposed by PARAFAC model according to the EEMs of DOM during AD process.

[23,27]. However, the mutual effects of NADH and the dissolved lignin-derived aromatics of lignocellulose commonly impact on the humic-like fluorescence peaks, resulting in an unexpected fluctuation of F_{max} scores. To avoid this condition, the ratio of just fulvic-like substances to protein-like matter is recommended [20]. In this study, two fluorescent components (C3 and C4) related to fulvic acid-like substrates were obtained. However, there was an obvious interference of protein-like matter on component C3 (Fig. 2), and therefore it was unsuitable for evaluating the biological stability of leachate samples. In contrast, for components C4 and C2 there was no disturbance to their individual ranges and they could be used as reliable monitoring indicators. The results of the C4/C2 ratios from all of the samples are shown in Fig. 3(c). As the AD proceeded from 28 to 84 days, the mean C4/C2 ratio in bioreactors R(8) and R(14) increased from 0.4 to 4.69, suggesting an increase in biological stability of DOM. For bioreactors AN and R(4), the C4/C2 ratios were maintained at a low level (less than 0.1). These results suggested, once again, that 8- and 14-day pre-aeration was effective for accelerating the stabilization of substrates in a dry AD system.

Zheng et al. [20] found a remarkable increase of protein-like matter due to excessive acidification during the wet AD of food waste. In another study by Lu et al. [38] in which starch wastewater was treated anaerobically, it was noted that the increase of fluorescence intensity related to protein-like substrates was coupled with the accumulation of VFAs. Generally, the results of this study were consistent with those mentioned above. Furthermore, this study confirms that, under the excessive acidification in the dry AD of MSW, the fluorescence DOM is notably dominated by protein-like materials, which could be considered reliable indicators to characterize the state of the excessive acidification in the reaction system. Additionally, for bioreactor R(8) the biogas generation almost ceased at 54 days (Fig. S4). It was also found that the F_{max} values of component C5 in 56- and 84-day leachate samples

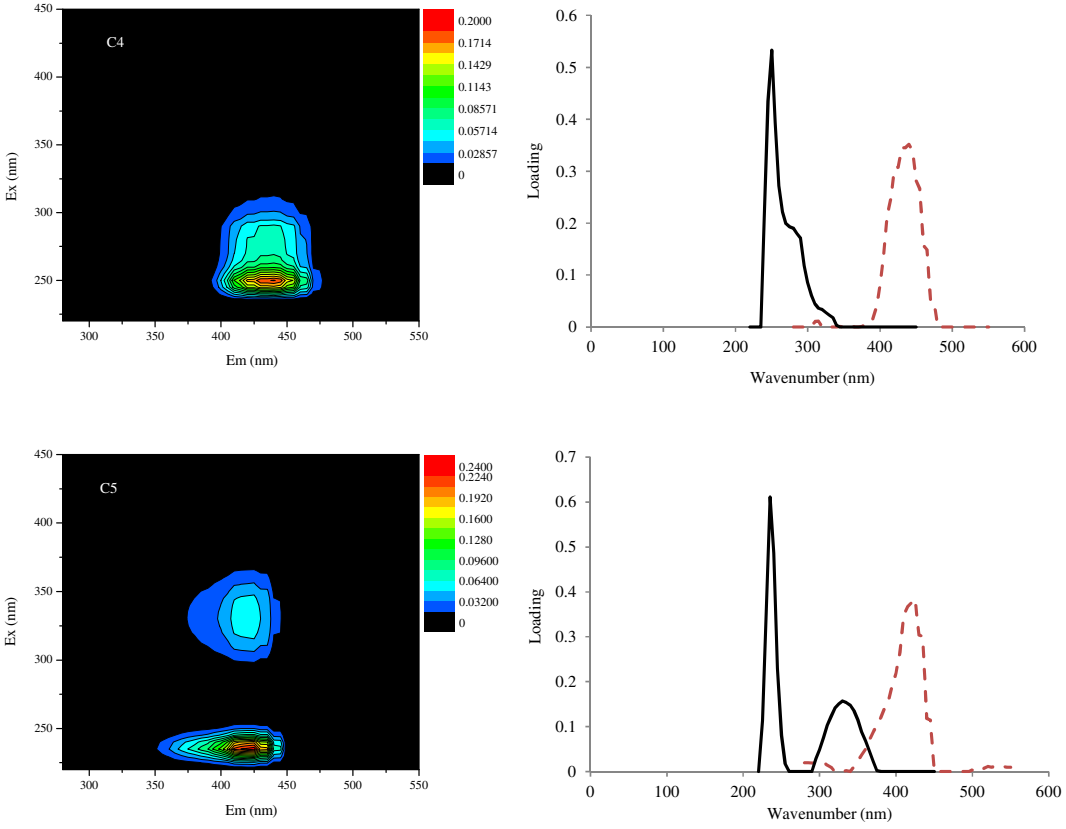


Fig. 2 (*continued*)

were not changed. Based on these findings, the potential correlations between bulk features and spectroscopic characteristics need further investigation.

3.5. Potential correlations between chemical and spectroscopic characteristics during the AD of pre-aerated MSW

Pearson correlation coefficients between spectroscopic parameters, including the F_{max} scores of five components from the EEM-PARAFAC and C4/C2 ratio, and the bulk characteristics of AD performance from all of the bioreactors over time (n = 12) are shown in Table 3. All of the spectroscopic parameters were correlated significantly ($P < 0.05$) with the main bulk parameters in leachate (e.g., pH, SCOD, and VFA content), which are usually used as indicators to distinguish the stages of AD [39]. Additionally, the spectroscopic parameters were associated with the cumulative biogas production rather than the biogas production rate, suggesting that the biodegradable spectroscopic components, such as protein-like matter, would be ultimately converted to biogas and stable DOM in leachate, but the biogas production rate was mutually influenced by biodegradable DOM concentrations, reactor conditions, and microbial activities.

Furthermore, PCA was used to evaluate the inter-relationships between the cases (leachate samples during AD from different bioreactors) and the variables (chemical and spectroscopic parameters). Four principal components were extracted, with the first two factors having a high loading for the variation, which accounted for ~94.2% of the total variance (81.6 and 12.6% for PC1 and PC2, respectively). The bi-plot graph for the PC1 and PC2 is shown in Fig. 4, and the cases (leachate samples) were also projected in the bi-dimensional space of the bi-plot. As shown in Fig. 4, all of the leachate samples from bioreactors AN and R(4) had similar chemical and spectroscopic characteristics during the AD process, and the concentrations of SCOD and VFAs were high due to the accumulation of protein-like matter (component C2) and the inhibition of stabilization processes. However, for leachate samples from bioreactors R(8) and R(14), as AD proceeded, the biodegradable organic matter was converted to stable material, resulting in a greater abundance of components C1, C4, and C5 in the middle and late term leachates, as well as more biogas production. Other parameters, such as the NH_4^+-N content and biogas generation rate, had a relatively weak relationship with PC1. Generally, the PCA results agreed with the analyses described in the previous sections, but PCA analysis was more intuitive for evaluation, as reported by Li et al. [24].

The results described above indicated that the spectroscopic characteristics from EEM-PARAFAC were promising indicators to characterize the performance of the dry AD of pre-aerated MSW.

3.6. Implications for pre-aeration prior to anaerobic treatment

Theoretically, a favorable methanogenic condition should balance the hydrolytic, acetogenic, and methanogenic activities and avoid potential inhibitions [10,11]. However, a dry anaerobic batch system filled with raw MSW with a large organic fraction usually suffers from excessive acidification due to the overload of organic matter, and therefore fails to establish a favorable methanogenesis. In this study, the organic loading in bioreactor AN was up to ~84 g VS/L, which is very high even for an AD with sufficient addition of inoculum [39]. Proteins and carbohydrates are considered to be the most easily hydrolyzed and acidified substrates during a typical AD process [20]. Pre-aeration for 4 days in bioreactor R(4) achieved a level of ~10.6% VS (accompanied by ~12.1% carbon) and carbohydrate losses of ~46.7%, but it had no obvious effect on relieving the excessive acidification and long lag-phase in the subsequent dry AD process (Figs. 1, S3 and S4). This could be explained

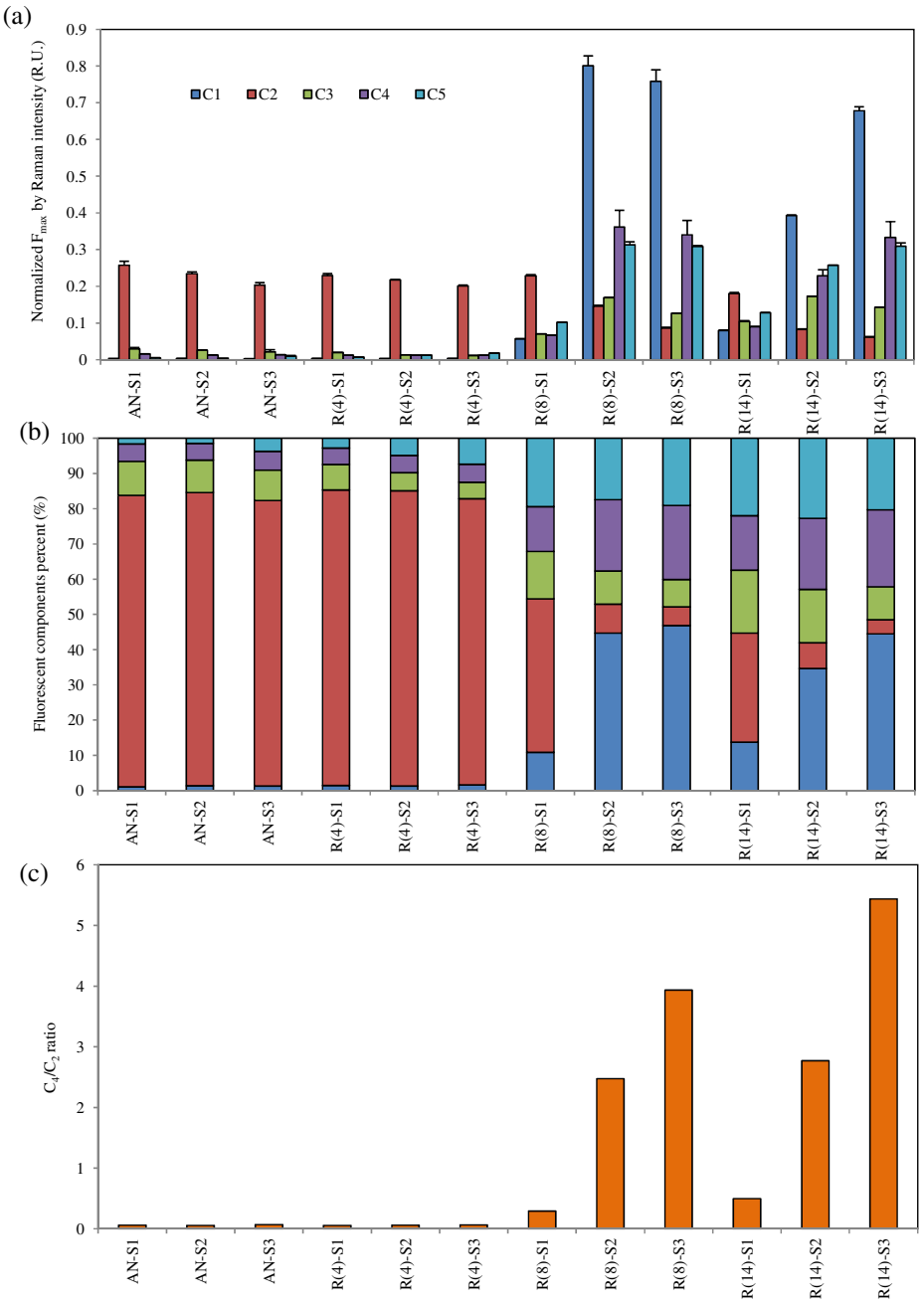


Fig. 3. F_{max} concentration scores (a) and relative percent (b) of different fluorescent components in leachate during AD process; C_4/C_2 ratio (c) (S1 represents 28-d samples; S2 represents 56-d samples; and S3 represents 84-d samples).

Table 3
Pearson correlation between spectroscopic and chemical parameters (n = 12).

	pH	SCOD	NH_4^+-N	AA	PA	iso-BA	n-BA	iso-VA	n-VA	TVFA	ABP	ABR
C1	0.786**	−0.766**	−0.429	−0.667*	−0.794**	−0.618*	−0.690*	−0.657*	−0.690*	−0.719**	0.951**	−0.257
C2	−0.766**	0.783**	0.525	0.639*	0.716**	0.545	0.698*	0.598*	0.708**	0.702*	−0.832**	0.232
C3	0.913**	−0.918**	−0.737**	−0.846**	−0.750**	−0.803**	−0.869**	−0.830**	−0.850**	−0.884**	0.939**	0.071
C4	0.842**	−0.832**	−0.525	−0.731**	−0.797**	−0.681*	−0.758**	−0.720**	−0.758**	−0.783**	0.971**	−0.176
C5	0.905**	−0.906**	−0.624*	−0.794**	−0.769**	−0.739**	−0.834**	−0.782**	−0.838**	−0.847**	0.977**	−0.050
C4/C2	0.736**	−0.740**	−0.534	−0.645*	−0.770**	−0.596*	−0.666*	−0.634*	−0.666*	−0.695*	0.853**	−0.261

AA, acetic acid; PA, propionic acid; iso-BA, iso-butyric acid; n-BA, *n*-butyric acid; iso-VA, iso-valeric acid; n-VA, n-valeric acid; TVFA, total volatile fatty acids; ABP, accumulated biogas production; ABR, biogas generation rate.

* Correlation is significant at the 0.05 level (2-tailed).

** Correlation is significant at the 0.01 level (2-tailed).

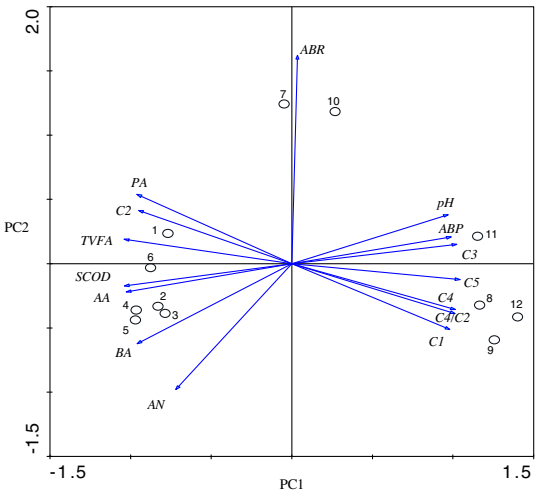


Fig. 4. Biplot illustrating the inter-relationships between and spectra parameters and bulk characteristics during dry AD of pre-aerated MSW (circle 1–3 represent 28-, 56- and 84-d leachate samples from AN; circle 4–6 represent 28-, 56- and 84-d leachate samples from R(4); circle 7–10 represent 28-, 56- and 84-d leachate samples from R(8); and circle 11–12 represent 28-, 56- and 84-d leachate samples from R(14); AA, acetic acid; PA, propionic acid; BA, butyric acid; TVFA, total volatile fatty acids; ABP, accumulated biogas production; ABR, biogas generation rate; AN, ammonia nitrogen).

by the 4-day pre-aeration only removing 6.6% of proteins during this time period, and therefore the majority of the protein content remained in the pre-treated waste. Moreover, EEM data for the AN and R(4) bioreactors showed that the accumulation of protein-like matter (component C2), which originated from the hydrolysis of the remaining proteins in aerobically pre-treated waste, occurred during the experimental period. Li et al. [24] reported that the tyrosine-like group in leachate that accumulated over time might become an inhibitory factor for converting organic matter to biogas. The accumulation of this group could hamper the hydrolysis of other organic polymers in the waste, resulting in the failure of the AD system.

The 8-day pre-aeration resulted in proteins losses of ~50.2%, with an accompanying carbohydrate removal of 78.8%. Therefore, the organic loading in R(8) dropped to ~62 g VS/L. Although this organic loading is still high during a typical AD process, no obvious inhibitions, as illustrated by the leachate parameters, were observed in bioreactor R(8), and its accumulated methane production reached 107.7 NL/kg DM. One possible cause of this, as observed in the initial leachate from bioreactor R(8), might be the presence of a suitable NH_4^+-N concentration at the beginning of the dry digestion process (Fig. S3). Zhai et al. [40] reported that an important factor impacting on pH was the NH_4^+-N concentration, which acts as a buffer under high VFA conditions in anaerobic leachate. Generally, the ammonification process is slow in a typical AD process [6,12], and therefore the unbalanced degradation rates of organic carbon and organic nitrogen tend to constrain microbial activity due to the lack of sufficient nutrition and the very low pH conditions in the liquid phase. In contrast, the degradation of organic nitrogen under aerobic conditions is much easier due to there being fewer limiting factors [41]. From this perspective, it could be concluded that suitable pre-aeration is capable of enhancing the ammoniation process of organic nitrogen in waste before anaerobic conditions arise, while the higher NH_4^+-N concentration in leachate could neutralize the excessive VFAs generated in the digesters to maintain an ideal pH for methanogenesis.

Based on the observations reported above, biological short-term pre-aeration could result in a different order of degradation of organic components in waste. Compared with carbohydrates, proteins are more resistant to aerobic biodegradation during this process and require a longer time to decompose. The degraded proteins could provide a certain amount of buffer capacity, and enhance the start-up of a subsequent dry AD. Therefore, the accelerated degradation of proteins in waste by pre-aeration is the most important mechanism that strengthens AD in a dry batch system. Previous studies [6,12] considered the use of pH values (6.5–8.0) and the VFA concentration (<6400 mg/L, in terms of COD) as monitoring parameters to determine the optimum switching point between in-situ aerobic pretreatment and anaerobic operation. Regarding the ex-situ pre-aeration process prior to anaerobic treatment (e.g., landfilling or dry AD), according to this study, a possible strategy for determining the switching point is to monitor the degree of degradation of proteins in waste, which is more reasonable for the avoidance of excessive pre-aeration than the use of bulk characteristics (e.g., VS and carbon losses) that have been reported in previous studies [17,18]. However, further experiments are needed to determine the optimum degree of proteins degradation to enhance methane production from the dry AD of MSW and reduce pre-treatment costs.

4. Conclusions

Short-term pre-aeration significantly improved the anaerobic degradation of MSW with a high organic fraction in a dry AD process. The accelerated degradation of proteins in waste by pre-aeration was a key factor in enhancing the start-up of a dry AD process. On one hand, protein degradation reduced the organic loading in a system, whereas on the other hand the ammonia generated provided a buffer to neutralize the excessive VFAs produced during the dry AD process. MSW with an 8-day pre-aeration obtained the highest biogas yield of 192.4 NL/kg DM, with an average methane content of 56.1% (v/v). This result represented a recovery of ~24.9% of the initial carbon in waste as methane during the integrated process of an 8-day pre-aeration and dry AD. When the pre-aeration was extended to 14 days, the cumulative biogas yield was ~33% lower than the 8-day pre-aeration due to the excessive organic matter consumption. To avoid excessive pre-aeration, it is advisable to interrupt the pre-aeration process based on the degree of proteins degradation and commence the subsequent anaerobic operation. Further investigation is needed to optimize this parameter by regulating pre-aeration modes.

In comparison, insufficient pre-aeration resulted in excessive butyrate-type acid inhibition during AD, with an accumulation of fluorescence components related to protein-like matter. Therefore, the response in the spectral properties of DOM (e.g., the relative abundance of C2 and the C4/C2 ratio) to the different reactor conditions indicates that these parameters can be used to identify process stability during the dry AD of MSW.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.cej.2016.11.020.

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